Fact finding techniques

Introduction

- Fact-finding is an important activity in system investigation. In this stage, the functioning of the system is to be understood by the system analyst to design the proposed system. Various methods are used for this and these are known as fact-finding techniques. The analyst needs to fully understand the current system.
- The analyst needs data about the requirements and demands of the project undertaken and the techniques employed to gather this data are known as fact-finding techniques.
- Various kinds of techniques are used and the most popular among them are interviews, questionnaires, record reviews, case tools and also the personal observations made by the analyst himself. Each of these techniques is further dealt with.

1. Interviews

- Interview is a very important data gathering technique as in this the analyst directly contacts system and the potential user of the proposed system.
- One very essential aspect of conducting the interview is that the interviewer should first establish a rapport with the interviewee . It should also be taken into account that the interviewee may or may not be a technician and the analyst should prefer to use day to day language instead of jargon and technical terms.
- For the interview to be a success the analyst should be appropriately prepared, as he needs to be beforehand aware of what exactly needs to be asked and to what depth. Also he should try to gather maximum relevant information and data. As the number and type of respondents vary, the analyst should be sensitive to their needs and nature.
- The advantage of interviews is that the analyst has a free hand and he can extract almost all the information from the concerned people but then as it is a very time consuming method, he should also employ other means such as questionnaires, record reviews, etc.

Interviews cont...d

- This may also help the analyst to verify and validate the information gained. Interviewing should be approached, as logically as possible and from a general point of view the following guides can be very beneficial for a successful interview.
  1. Set the stage for the interview.
  2. Establish rapport; put the interviewee at ease.
  3. Phrase questions clearly and succinctly.
  4. Be a good listener; avoid arguments.
  5. Evaluate the outcome of the interview.
- The interviews are of two types namely structured and unstructure

Structured Interview

- Structured interviews are those where the interviewee is asked a standard set of questions in a particular order. All interviewees are asked the same set of questions.

Interviews cont...d

- The questions are further divided in two kinds of formats for conducting this type of interview.
- The first is the open-response format in which the respondent is free to answer in his own words. An example of open-response is "Why are you dissatisfied with the current leave processing method?"
- The other option is of closed-response format, which limits the respondents to opt their answer from a set of already prescribed choices. An example of such a question might be "Are you satisfied with the current leave processing methods?" or "Do you think that the manual leave processing procedure be changed with some automated procedure?"

Unstructured Interview

- The unstructured interviews are undertaken in a question-and-answer format. This is of a much more flexible nature than the structured interview and can be very rightly used to gather general information about the system. Here the respondents are free to answer in their own words. In this way their views are not restricted. So the interviewer get a bigger area to further explore the issues pertaining to a problem.

2. Questionnaires

- Questionnaires are another way of information gathering where the potential users of the system are given questionnaires to be filled up and returned to the analyst.
- Questionnaires are useful when the analyst need to gather information from a large number of people. It is not possible to interview each individual. Also if the time is very short, in that case also questionnaires are useful. If the anonymity of the respondent is guaranteed by the analyst then the respondent answers the questionnaires very honestly and critically.
- The questionnaire may not yield the results from those respondents who are busy or who may not give it a reasonable priority.
- The analyst should sensibly design and frame questionnaires with clarity of it's objective so as to do justice to the cost incurred on their development and distribution. Just like the interviews and on the same lines questionnaires are of two types i.e. *Open-Response Based* and the *Closed-Response Based*

Questionnaires cont...d

Open-Response Based Questionnaires

- The objective of open-response questionnaire is to gather information and data about the essential and critical design features of the system. The open-ended question requires no response direction or specific response.
- This form is also used to learn about the feelings, opinions, and experiences of the respondents. This information helps in the making the system effective because the analyst can offer subsequent modifications as per the knowledge gained.

Closed-Response Based Questionnaires

- The objective of closed-response questionnaire is to collect the factual information of the system. It gives an insight in how the people dealing with the system behave and how comfortable are they with it. In this case the respondents have to choose from a set of given responses. Thus the respondent can express their liking for the most favorable one from the possible alternatives.
- The closed questions can be of various types and the most common ones are listed below.
  1. Fill-in-the-blanks.
  2. Dichotomous i.e. Yes or No type.

Questionnaires cont...d

3. Ranking scale questions ask the respondents to rank a list of items in the order of importance or preference.

4. Multiple-choice questions which offer respondents few fixed alternatives to choose from.

5. Rating scale questions are an extension of the multiple-choice questions. Here the respondent is asked to rate certain alternatives on some given scale.

Open Response-Based Vs Closed Response-Based

- The basic comparison between the two can be made on the grounds of the format used. Open form offers more flexibility and freedom to the respondents whereas the closed form is more specific in nature.
- Open-ended questionnaires are useful when it is required to explore certain situation. They also require a lot of time of the analyst for evaluation.
- Closed questionnaires are used when factual information is required. Closed questions are quick to analyze but typically most costly to prepare but they are more suitable to obtain factual and common information.
- It is the job of the analyst to decide which format should be employed and what exactly should be it's objective. The care should be taken to ensure that all the parts of the form are easy to understand for the respondents so that they can answer with clarity

3. Record Reviews

- Records and reports are the collection of information and data accumulated over the time by the users about the system and it's operations. This can also put light on the requirements of the system and the modifications it has undergone. Records and reports may have a limitation if they are not up-to-date or if some essential links are missing. All the changes, which the system suffers, may not be recorded.
- The analyst may scrutinize the records either at the beginning of his study which may give him a fair introduction about the system and will make him familiar with it or in the end which will provide the analyst with a comparison between what exactly is/was desired from the system and it's current working.
- One drawback of using this method for gathering information is that practically the functioning of the systems is generally different from the procedure shown in records. So analyst should be careful in gathering in gathering information using this method.

4. On-site Observation

- On-site observations are one of the most effective tools with the analyst where the analyst personally goes to the site and discovers the functioning of the system. As an observer, the analyst can gain first hand knowledge of the activities, operations, processes of the system on-site, hence here the role of an analyst is of an information seeker.
- This information is very meaningful as it is unbiased and has been directly taken by the analyst. This exposure also sheds some light on the actual happenings of the system as compared to what has already been documented, thus the analyst gets closer to the system. This technique is also time-consuming and the analyst should not jump to conclusions or draw inferences from small samples of observation rather the analyst should be more patient in gathering the information. This method is however less effective for learning about people's perceptions, feelings and motivations.

Fact Finding Techniques - Self Assessment

1. What are the two types of interviews used in fact finding techniques?

2. How many types of questionnaires are used? Compare them. State True or False

1. In open response questionnaires, the respondent is free to answer in his/her own words.

2. Decision tree is two dimensional matrix showing different conditions and their respective actions taken.

Fact Finding Techniques - Summary

Key points covered in this chapter are:

" Fact finding techniques are very useful during the analysis phase. There are many fact finding techniques avalable like interviews, questionnaires, on site observation etc.

" Each of the technique has its own format, suitability, merits and demerits. The analyst has to make a judgement on which format to be used to collect what information and who should be the respondents.

" Decision-making is an integral part of any business setup. To facilitate the decision making process, which is based on the possibilities of various conditions and their combination there, arises a need to document them that will make the whole process easier.

" First step in documenting is the identification of the systems conditions and actions. Then the available tools are used. The three tools available are decision trees, decision tables and Structured English.

" In the decision trees the decision variables are represented in a tree's branch structure. The conditions are depicted sequentially and the root of the tree is the starting point.

" Decision tables links the conditions and actions by using decision rules, which state all the conditions that must be true for a particular set of actions, are taken.

" Structured English is employed to state decision rules and it has three kinds of statements namely sequence, decision and iteration structures. This format is very concise and compact and also very easy to understand.

Fact Finding Techniques - Exercises

1. What type information is often best obtained through interview?

2. Which method does an analyst use to gather data?

3. What is fact-finding?

4. Make a list of dos and don'ts that you should follow when interviewing a user.

5. List diffèrent types of questionnaire formats.

6. What are the differences between open and closed-form questionnaires? For what types of information is each form most useful?

7. When is record review an effective data collection method? What is the purpose of this method?

8. What role does observation play in system investigation?

9. What are conditions and actions? What are their roles in decision analysis?

10. In what way do decision trees assist in decision analysis? Explain how an analyst should develop a decision tree.

11. What advantages do decision trees present for analysis?

12. How does the purpose of decision tables differ from that of decision trees? What components make up a decision tables?

13. How do analysts develop a decision table?

14. How a decision table is developed?

15. How does structured English differ from the decision tree and decision table? What advantages does it offer over the other two methods?

16. Summarize the advantages and limitations of interviews and questionnaires.

17. Explain the differences between

(a) structured and unstructured interviewing and

(b) open-ended and closed questions. Give an example of each.